Scalability & Future Plan

Date	30 SEPTEMBER 2026
Team ID	SWTID-2026-1660
Project Name	AI WEATHERWISE API
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Depends on external Weather API	System may not work properly if the API is unavailable	High
2	Limited to basic weather information	Users may need more advanced weather features	Medium
3	Limited handling of large numbers of users	Performance may decrease with many users	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports normal number of users	Use cloud hosting and load balancing for more users
2	Data Storage	Limited data storage	Use a scalable database for storing weather history
3	Performance	Suitable for normal usage	Use caching and API optimization to improve response time
4	Security	Basic security measures	Add authentication, API key protection and HTTPS

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact

Phase 2	Weather forecast and weather history	1 – 2 Months	Provides more useful weather information
Phase 3	AI-based weather prediction and personalized alerts	3 – 4 Months	Improves prediction and user experience
Phase 4	Mobile application and cloud deployment	5 – 6 Months	Makes the system accessible to more users